



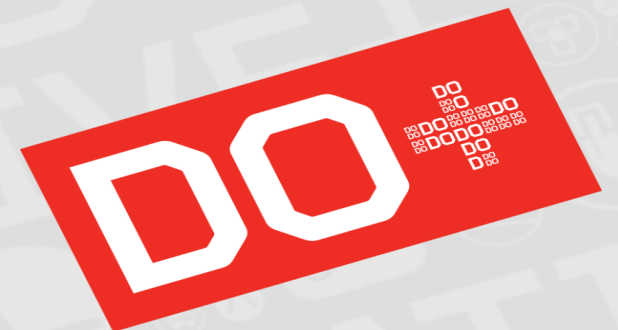
EBG Server Education: Resource Discovery, Management, Inventory

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lenovo.

XClarity™ Administrator

Version 1.0.0



- Resource Discovery
 - Automatic SLP Discovery
 - Manual Discovery
- Resource Management
- Managed Devices Inventory
- Unmanaging Resources

Resource Discovery

- Automatic SLP Discovery

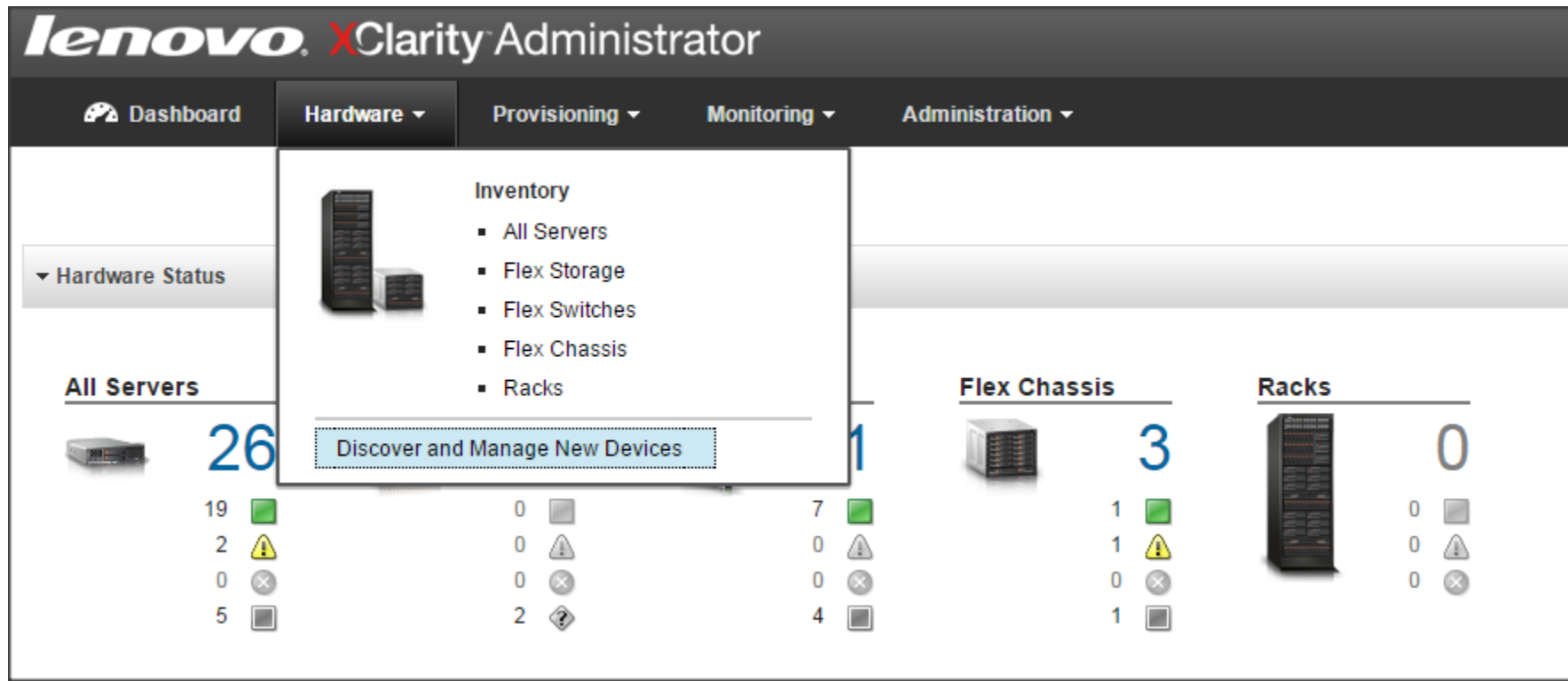
- LXCA periodically looks for compatible hardware
- Compatible hardware is displayed in the discovered hardware grid
- Supported endpoints can be selected for management from that grid
- Managed resources do not appear in the grid
- Can be turned off by deselecting the “Probe for Manageable Systems” option

- Manual Discovery

- One or more endpoints can be discovered during the “Manual Input” and “Bulk Import” manage operations
- Allows for non-SLP discoverable systems to still be managed
- Non-compatible endpoints will fail the manage operation

Navigating to Resource Discovery Page

- Select the Hardware Menu
- Select the “Discover and Manage New Devices” sub-menu



Discover and Manage New Devices Page

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Status Jobs RAJ ?

Dashboard Hardware Provisioning Monitoring Administration

Discover and Manage New Devices

+ Manual Input

+ Bulk Import

☒ Probe for Manageable Systems

 | Manage Selected |

Filter

☐	Name	IP Addresses	Serial Number	Type	Type-Model	Manage Status
☐	SN#Y031BG23200J	10.243.2.80, fe80::5ef3:f...	23DWN55	Chassis	8721-HC1	Ready
☐	PIT-CMM-150	10.243.0.141, fe80::42f...	23DZX85	Chassis	8721-HC1	Ready
☐	SN#Y010BG3C902S	10.243.3.191, fe80::42f...	06YYML3	Chassis	8721-HC1	Ready
☐	SN#Y013BG25P0RA	10.243.6.200, fe80::364...	23DWN55	Chassis	8721-HC1	Ready
☐	SN#Y031BG23200B	10.243.3.43, fe80::5ef3:f...	23DWN55	Chassis	8721-HC1	Ready
☐	SN#Y010BG48D07C	10.243.7.151, fe80::42f...	06CAVWN	Chassis	8721-HC1	Ready
☐	SN#Y010BG48D01J	10.243.7.165, fe80::42f...	06CAVWP	Chassis	8721-HC1	Ready
☐	SN#Y010BG35G03K	10.243.0.228, fe80::364...	71D1039	Chassis	8721-CTO	Ready

Resource Management

- The manage operation
 - Puts the endpoint into system inventory
 - Configures the endpoint to centralized authentication
 - Configures the endpoint to the setup NTP server
 - Configures the endpoint to forward events and audit events to LXCA

Resource Management

- Managing SLP Discovered Systems
 - Select one or more supported endpoints from the Manageable Systems grid
 - Press the “Manage Selected” button

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Dashboard Hardware Provisioning Monitoring Administration

Discover and Manage New Devices

+ Manual Input + Bulk Import

☒ Probe for Manageable Systems

Manage Selected

☐	Name	IP Addresses	Serial Number	Type	Type-Model	Manage Status
☒	SN#Y031BG23200J	10.243.2.80, fe80::5ef3.f...	23DWN55	Chassis	8721-HC1	Ready
☐	PIT-CMM-150	10.243.0.141, fe80::42f...	23DZX85	Chassis	8721-HC1	Ready

Resource Management: Managing SLP Discovered System

- Fill in the required fields and select “Manage”

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Dashboard Hardware Provisioning Monitoring Administration

Discover and Manage New Devices

+ Manual Input + Bulk Import

☒ Probe for Manageable Systems

Manage Selected

Name	IP Address	Manage Status
☒ SN#Y031BG23200J	10.243.2.1	Ready
☐ PIT-CMM-150	10.243.0.1	Ready
☐ SN#Y010BG3C902S	10.243.3.1	Ready
☐ SN#Y013BG25P0RA	10.243.6.1	Ready
☐ SN#Y031BG23200B	10.243.3.2	Ready
☐ SN#Y010BG48D07C	10.243.7.1	Ready

Manage

i SN#Y031BG23200J is going to be managed.

Specify chassis management module (CMM) credentials

user name (supervisor name) ✖ This value is required.

password

Specify the password for the CMM recovery ID ?

RECOVERY_ID password

confirm RECOVERY_ID password

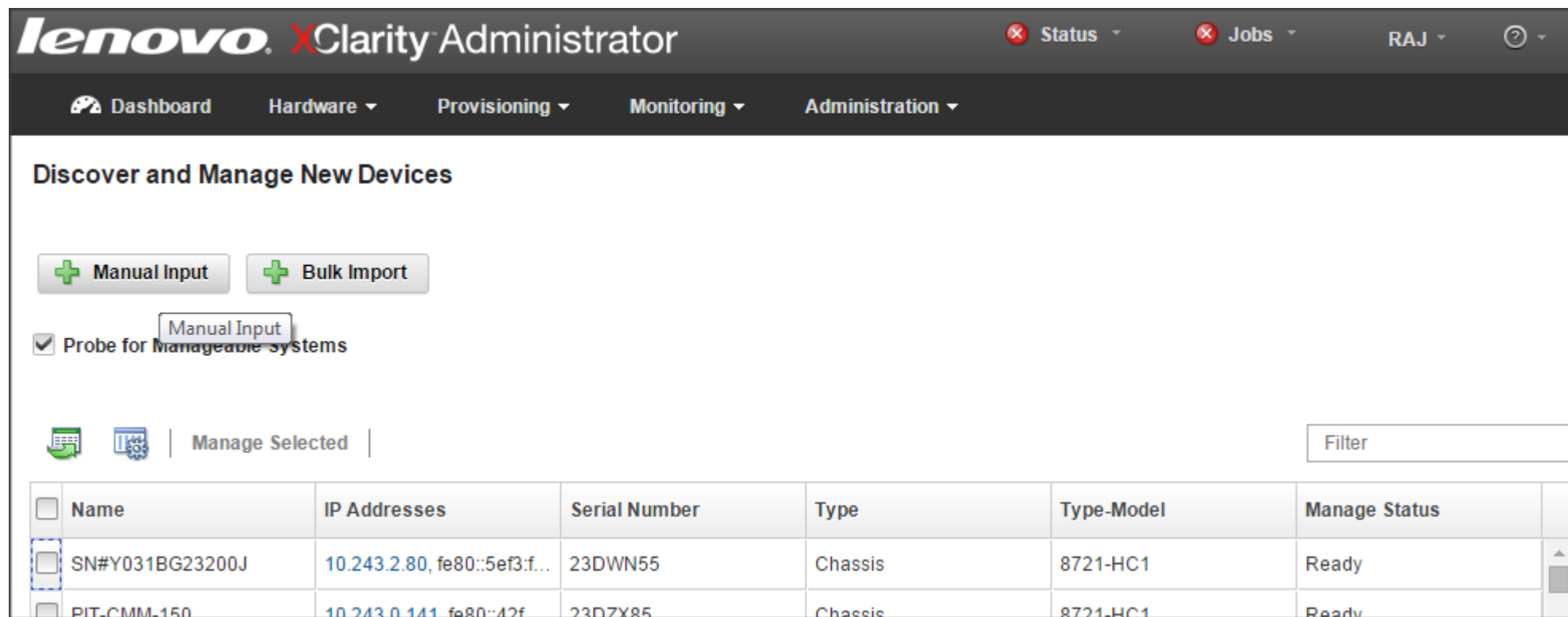
This will perform the following actions against chassis:

1. Configure NTP clients to use the NTP settings from Lenovo® XClarity Administrator
2. Configure for centralized user management

Manage Cancel

Resource Management

- Managing via Manual Input
 - Select the “Manual Input” button



The screenshot displays the Lenovo XClarity Administrator web interface. The top navigation bar includes the 'lenovo. XClarity Administrator' logo and user information (Status, Jobs, RAJ). The main menu shows 'Dashboard', 'Hardware', 'Provisioning', 'Monitoring', and 'Administration'. The 'Discover and Manage New Devices' section is active, featuring buttons for '+ Manual Input' and '+ Bulk Import'. A checkbox labeled 'Probe for Manageable Systems' is checked, with a tooltip 'Manual Input' appearing over it. Below this, there are icons for a calendar and a gear, followed by the text 'Manage Selected' and a 'Filter' input field. A table lists the discovered devices:

	Name	IP Addresses	Serial Number	Type	Type-Model	Manage Status
☐	SN#Y031BG23200J	10.243.2.80, fe80::5ef3:f...	23DWN55	Chassis	8721-HC1	Ready
☐	PIT-CMM-150	10.243.0.141, fe80::42f...	23DZX85	Chassis	8721-HC1	Ready

Resource Management: Manual Input

- Select “Single System” or “Multiple Systems” radio button

The screenshot displays the Lenovo XClarity Administrator web interface. A modal dialog box titled "Manual Input" is centered on the screen. The dialog has two radio buttons: "Single System" (which is selected) and "Multiple Systems". Below these is a text input field labeled "IP Address:". At the bottom of the dialog are "OK" and "Cancel" buttons. In the background, the main interface shows a "Discover and Manage New Devices" section with buttons for "Manual Input" and "Bulk Import", a checked checkbox for "Probe for Manageable Systems", and a table of discovered systems.

Name	IP Address
SN#Y031BG23200J	10.243.2
PIT-CMM-150	10.243.0
SN#Y010BG3C902S	10.243.3
SN#Y013BG25P0RA	10.243.6
SN#Y031BG23200B	10.243.3
SN#Y010BG48D07C	10.243.7
SN#Y010BG48D01J	10.243.7

Manage Status
Ready
Ready
Ready
Ready
Ready
Ready
Ready

Resource Management: Manual Input – Single System

- Enter the IP of the single targeted system and select “OK”

lenovo. XClarity Administrator

Dashboard Hardware

Discover and Manage New Devices

+ Manual Input + Bulk Import

☒ Probe for Manageable Systems

Manage Selected

Name	IP Address
☐ SN#Y031BG23200J	10.243.2...
☐ PIT-CMM-150	10.243.0...
☐ SN#Y010BG3C902S	10.243.3...
☐ SN#Y013BG25P0RA	10.243.6...
☐ SN#Y031BG23200B	10.243.3...
☐ SN#Y010BG48D07C	10.243.7...
☐ SN#Y010BG48D01J	10.243.7...

Manual Input

☒ Single System ☐ Multiple Systems

IP Address: 10.243.3.61

OK Cancel

Filter

Manage Status
Ready
Ready
Ready
Ready
Ready
Ready
Ready
Ready

Resource Management: Manual Input – Multiple Systems

- Enter the IP ranges of the target systems and select “OK”

Manual Input

☐ Single System ☒ Multiple Systems

10.243.3.61 - 10.243.3.63 ✖

10.243.3.71 - 10.243.3.73 ✖+

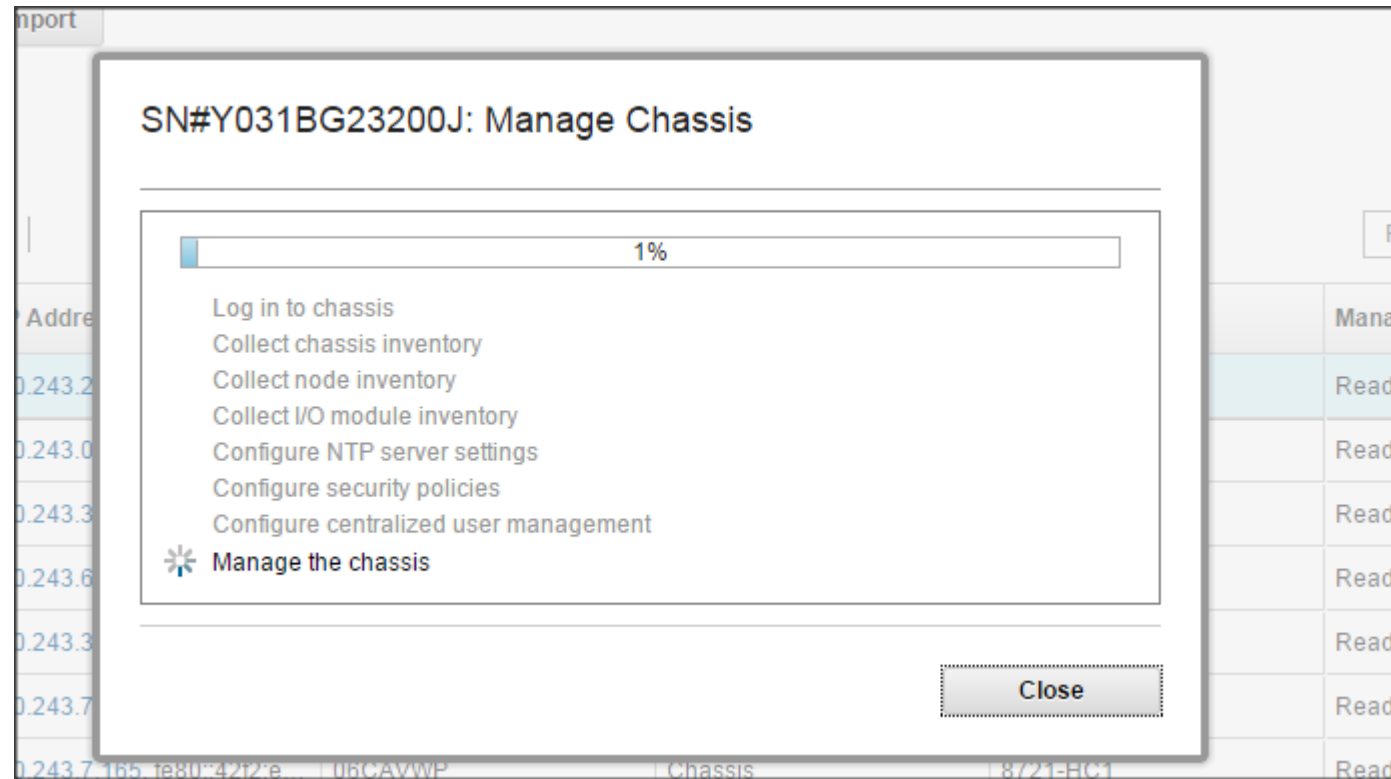
6 addresses to be added.

OK Cancel

Name	IP Address	Manage Status
SN#Y031BG23200J	10.243.2...	Ready
PIT-CMM-150	10.243.0...	Ready
SN#Y010BG3C902S	10.243.3...	Ready
SN#Y013BG25P0RA	10.243.6...	Ready
SN#Y031BG23200B	10.243.3...	Ready
SN#Y010BG48D07C	10.243.7...	Ready
SN#Y010BG48D01J	10.243.7...	Ready

Resource Management: Manage Dialogue

- Wait for manage dialogue to complete
- Optionally close the dialogue for the manage operation to run as a background job



Resource Management: Potential Errors

■ Potential Manage Errors:

Manage task:

- 1) Validate network address and connectivity
 - Fail if unable to connect or unable to determine management IP address
 - Verify IP address, account credentials, slp response
- 2) Determine if already managed by either xClarity or FSM
 - Fail if a CIM subscription from one of those products is already configured on the endpoint
 - Manually remove the subscription from the endpoint
- 3) Gather initial inventory
 - This could fail for TCP command mode or CIM errors that must first be resolved
 - Verify endpoint is configured to allow these protocols
- 4) Verify interoperability
 - Fail if endpoint is not supported
- 5) Update DNS settings
 - Does not fail management if this fails
- 6) Update NTP settings and sync time
 - Does not fail management if this fails
- 7) Retrieve endpoint certificate(s) needed
 - Will fail if unable to retrieve appropriate certificates and add to trust store (CMM CA certificate, rack server LDAP and CIM certificates)

Potential Manage Errors (cont)

Manage Task (cont)

- 8) Update connections to make use of certificate(s)
Does not fail management
- 9) Install xClarity certificate on endpoint LDAP service
Fail management if unable to install certificate on endpoint
Review endpoint event logs and xClarity logs for errors
- 10) Install CIM indication subscription on endpoint
Fail management
Review endpoint and xClarity logs for errors
- 11) Configure LDAP settings
Fail management
Review endpoint and xClarity logs for errors
- 12) Update connections to make use of xClarity user accounts
Fail management
Review endpoint and xClarity logs for errors
- 13) Enable centralized management
Fail management
Review endpoint and xClarity logs for errors

Resource Inventory

- Managed resource inventory is provided throughout the LXCA user interface
- Provided inventory includes:
 - System description information (MTM/SN/UUID)
 - Installed components
 - System firmware levels
 - Health status
 - System LEDs
- LXCA refreshes resource inventory on event and on periodic intervals

Unmanaging Resources

- The unmanage operation
 - Removes the endpoint from system inventory
 - Removes the centralized authentication configuration from the endpoint
 - Removes the NTP configuration from the endpoint
 - Removes event forwarding from the endpoint

Unmanaging Resources

- Navigate to an endpoint hardware grid (Chassis or Rack)
- Select one or more endpoint to unmanage
- Select the “Unmanage” button

lenovo. XClarity Administrator ✖ Status ✖ Jobs RAJ ?

Dashboard Hardware Provisioning Monitoring Administration

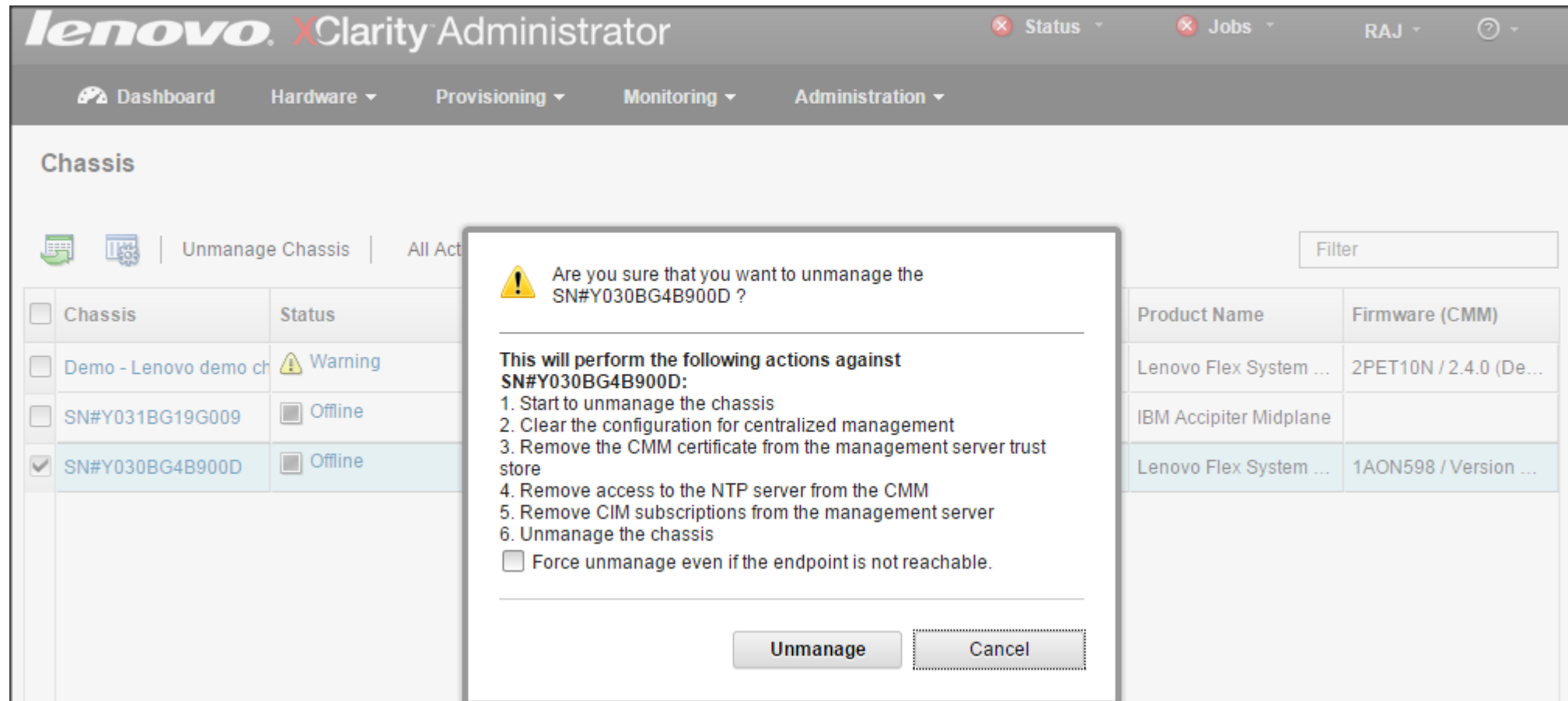
Chassis

  Unmanage Chassis | All Actions Filter

☐	Chassis	Status	IP Addresses	Type-Model	Serial Number	Product Name	Firmware (CMM)
☐	Demo - Lenovo demo ch	 Warning	169.254.1.3, 169.25...	8721-HC1	00001X8721	Lenovo Flex System ...	2PET10N / 2.4.0 (De...
☐	SN#Y031BG19G009	 Offline		8721-FT3	B080SPb	IBM Accipiter Midplane	
☒	SN#Y030BG4B900D	 Offline	10.243.3.61, 10.243....	8721-HC1	23DGL39	Lenovo Flex System ...	1AON598 / Version ...

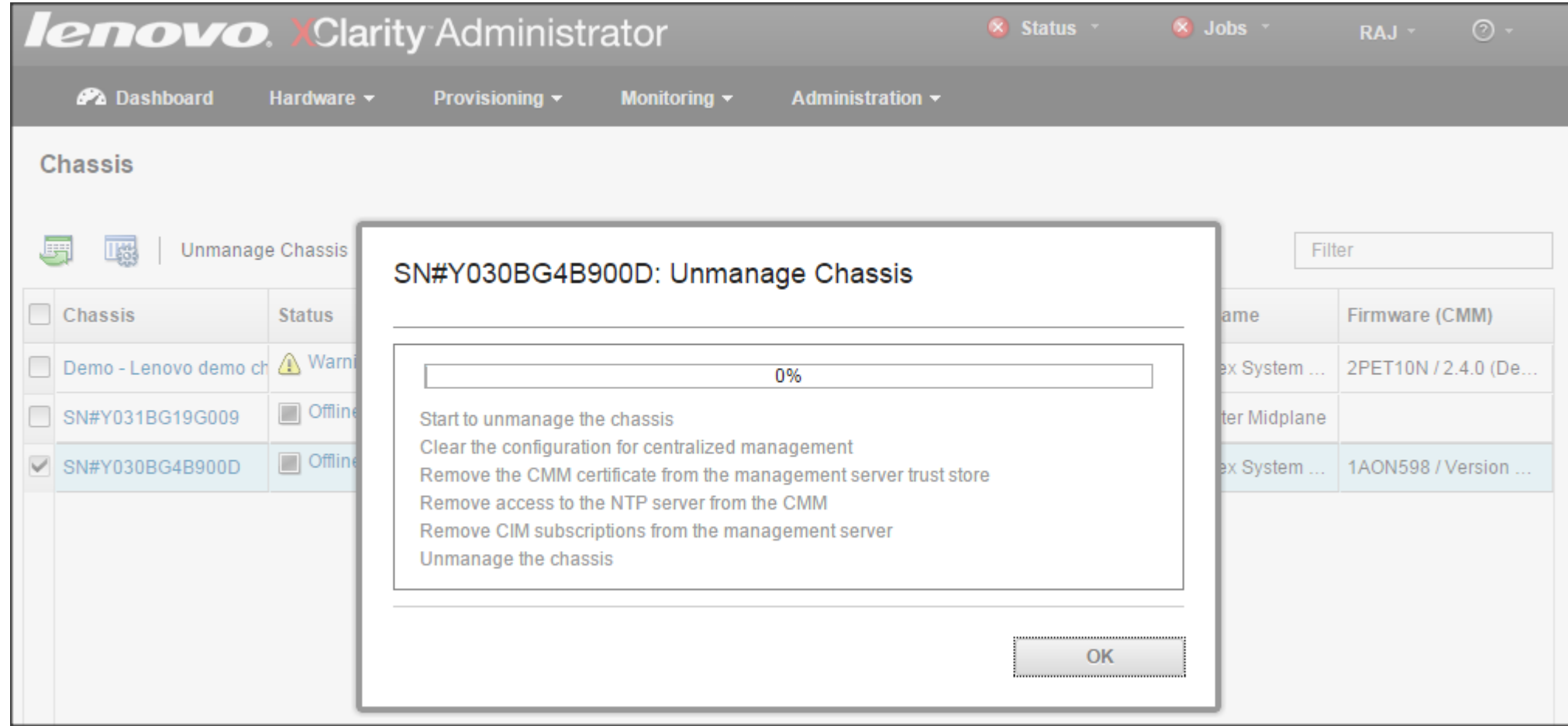
Unmanaging Resources

- Confirm the unmanage operation
 - Optionally force the operation to unmanage unreachable endpoints



Unmanaging Resources

- Wait for the unmanage progress dialogue to complete



Potential UnManage Errors

UnManage Task:

If for any reason you can not unmanage an endpoint, you can select the 'Force' option. But the endpoint may need to be cleaned up to allow management again (i.e. clear CIM subscriptions)

1) Connectivity check to the endpoint

If we are unable to communicate with the endpoint management IP address we will fail

Use the force option

2) Disable NTP settings

Will not fail the unmanage task

3) Remove CIM subscription

Will not fail the unmanage task, may require additional cleanup later

4) Delete EHCM node account on chassis

Will not fail the unmanage task

5) Disable centralized management

If we fail this we will fail the unmanage task

Most typical failure is a timeout on the disable centralized management on a CMM. Try again or clean up endpoint manually and do a force option in xClarity.

6) Remove endpoint certificate(s) from xClarity trust store

Will not fail the unmanage task

THANK YOU GRACIAS
MERCI DANKE GRAZIAS 謝謝
OBRIGADO ありがとう DANK TAKK BEDANKT СПАСИБО
DAKUJEM